SULI Internship: Extreme Weather Analysis

National Geographic, 2020





Melody Nguyen

Summer 2024

SLAC National Accelerator Laboratory is operated for the US Department of Energy by Stanford University

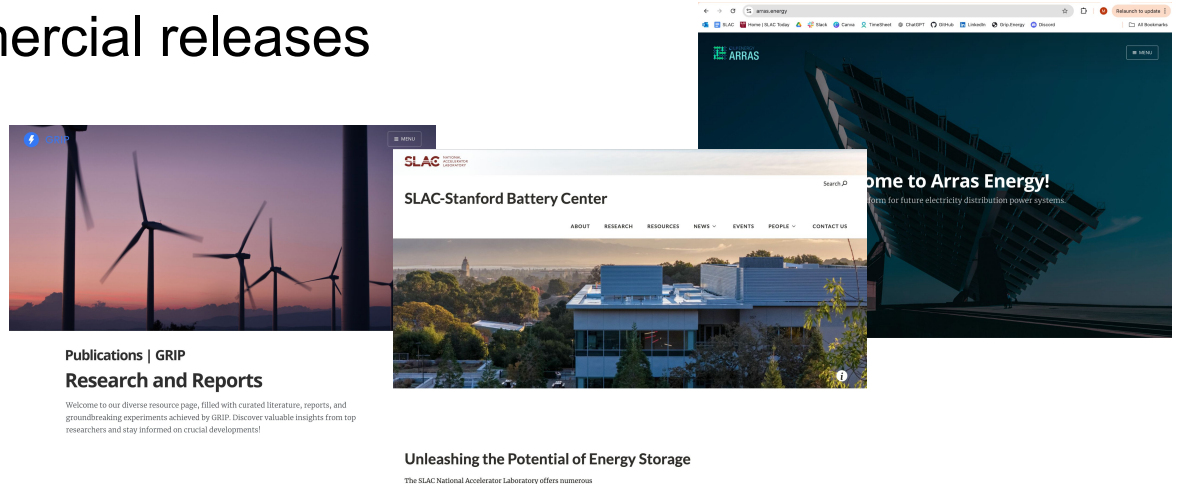
U.S. DEPARTMENT OF ENERGY
Office of Science

GISMO Grid Integration, Systems & Mobility

SLAC NATIONAL ACCELERATOR LABORATORY

Background

- Applied Energy Division
  - Grid Integration Systems & Mobility (GISMo)
- Marketing and Analytics
  - Web development: HTML, CSS, markdown
  - Launch of commercial releases
- Inspired to code!



Background

- Climate Change
  - Rise in temperatures, “abnormal” conditions
  - Increasing frequency & intensity of extreme weather
- Challenge: Overload on electrical grid
  - High demand of energy
  - Insufficient load supply
  - Forced rolling blackouts
  - Wildfires



Report: Root Cause Analysis
Mid-August 2020 Extreme Heat Wave

Background

- How can we improve **resilience of the electrical grid**, and maintenance crews to be ready for these events?
- Can **renewable energy** resources help supply energy loads?





IEEE JOURNAL OF PHOTOVOLTAICS, NOVEMBER 2023

Background

- **Photovoltaic (PV)** generators
  - Correlated to spatial averaging of local weather conditions
- **Solar Industry**
  - Less variability, more stable fleet behavior
- Extreme weather events
  - **High temperature**
  - Fire smoke → reduced irradiance
- Loss of PV generation



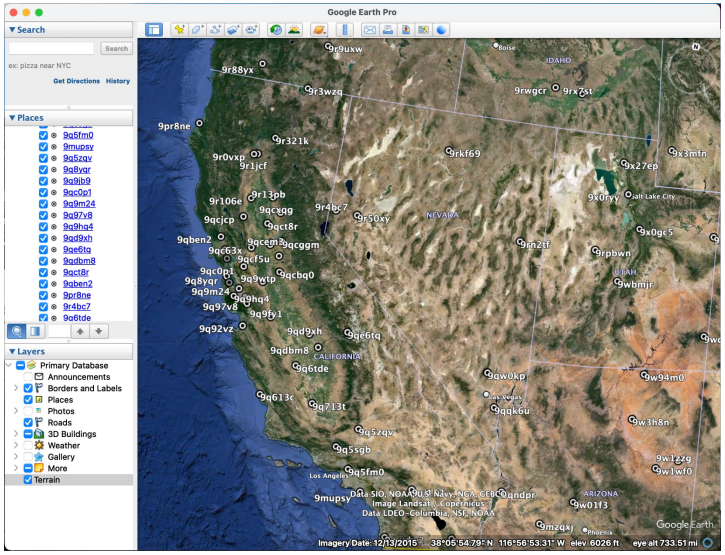
REGROW TECHNICAL WORK PLAN
Meyers, B and Perry, K (February, 2024)

Background

- **REGROW**
  - Renewable Energy Generation Risk from Outlier Weather
  - Studies impact of extreme weather events on performance of PV systems
  - Statistical model of fleet power production (WECC)
- **GRIP**
  - Grid Resilience & Intelligence Platform
  - Software to quantify grid status and pole vulnerability
  - Provide actionable insights and solutions to utility operators

Motivation

- **Case Study**
  - 2020 August heatwave
  - Forecasting of abnormality
  - Locations across US west coast
- **Internship Goal**
  - Data analytics
  - How to identify extreme event?
  - *Predict* future heat waves to better operate PV generators

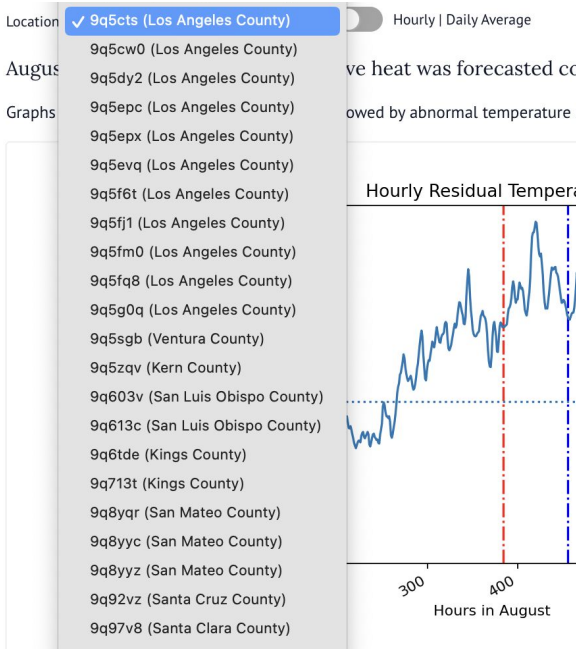


Methodology

- **Python Programming**
  - Marimo notebook
  - Interactive UI features
  - Graphics

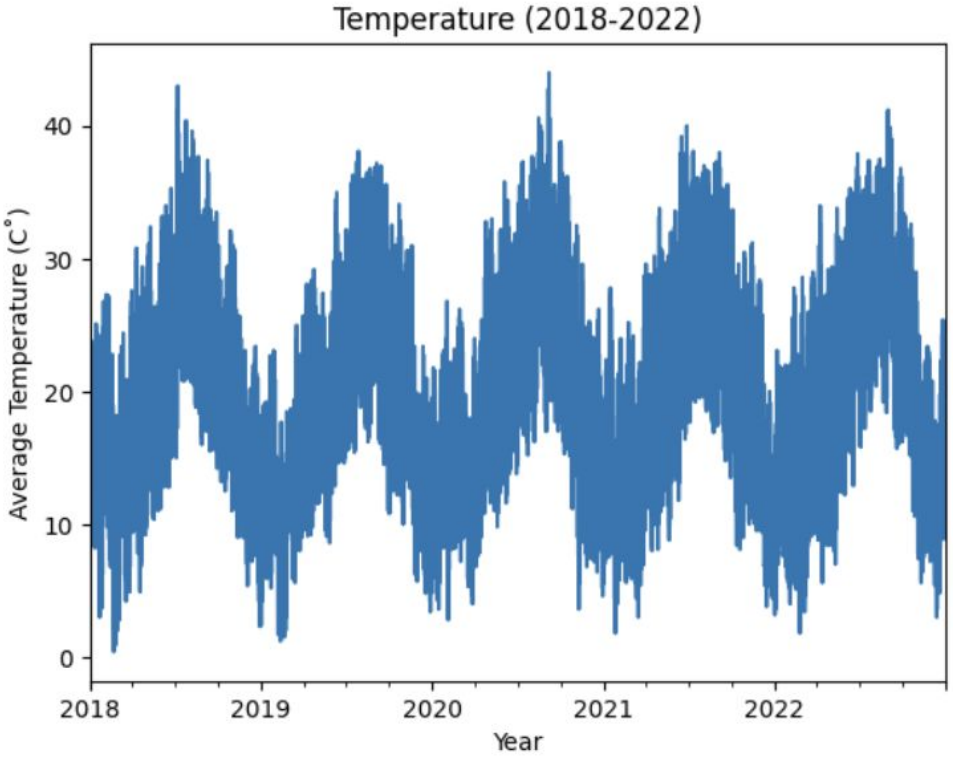
Select a node: 9mupsy

```
# Drop down selection for all nodes, default selection is the first node
nodes_dropdown = mo.ui.dropdown(nodes, value=nodes[0], label='Select a node:')
nodes_dropdown
```



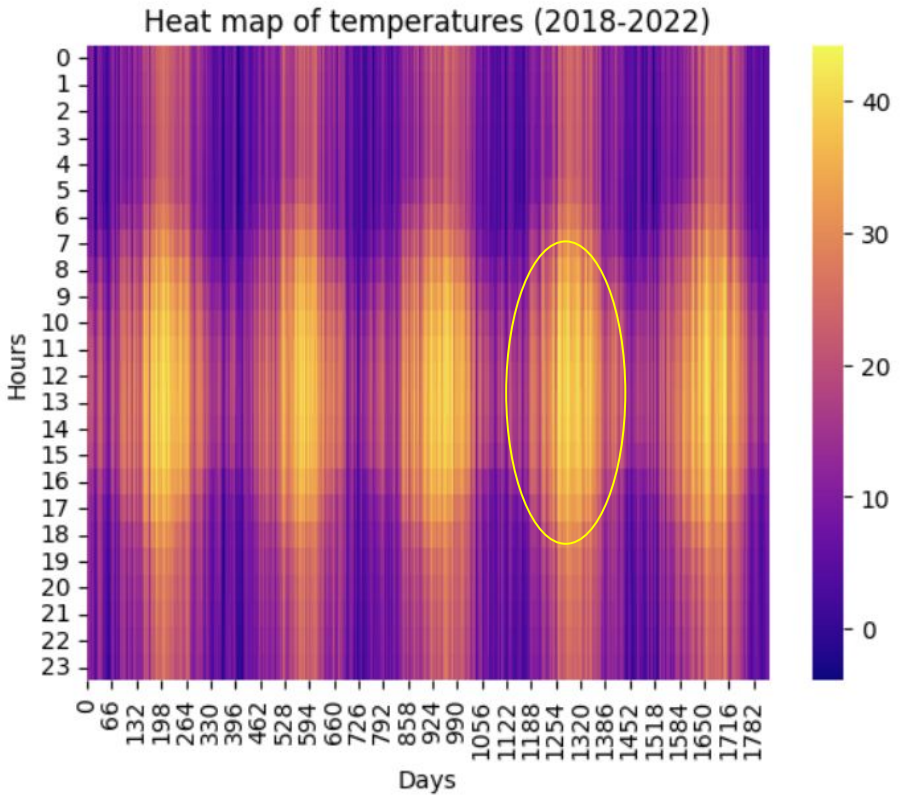
Experimenting

- **Graphing Data**
- Initial visualization
  - Linear graph
  - Over five years
- *Los Angeles, CA*
  - *Node: 9q5epc*



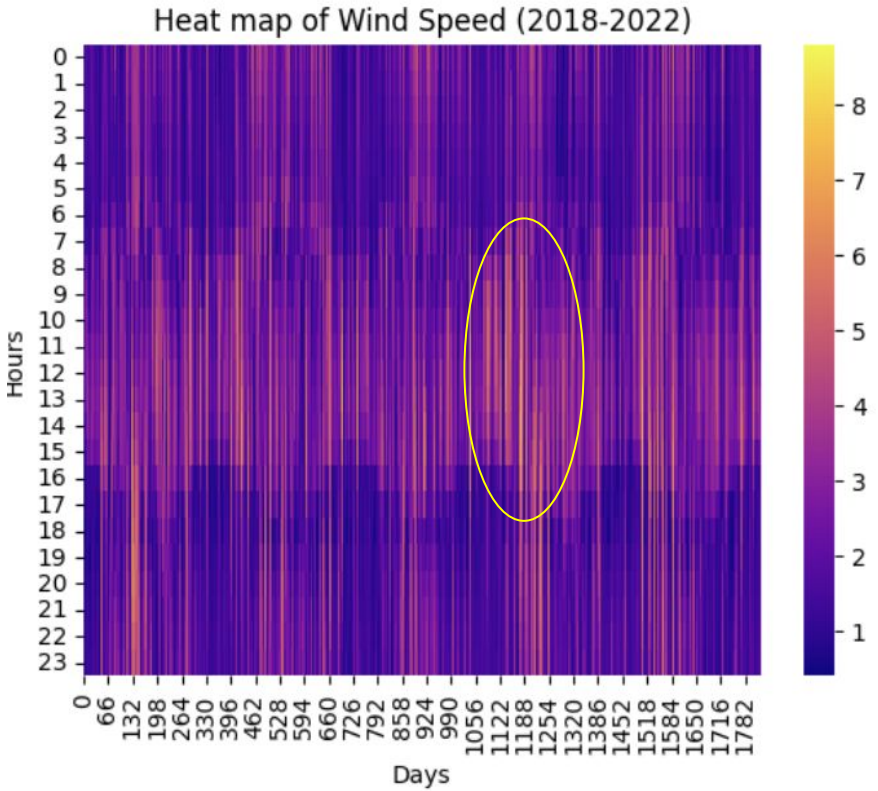
Experimenting

- **Graphing Data**
  - Observing intensity
  - Heat map

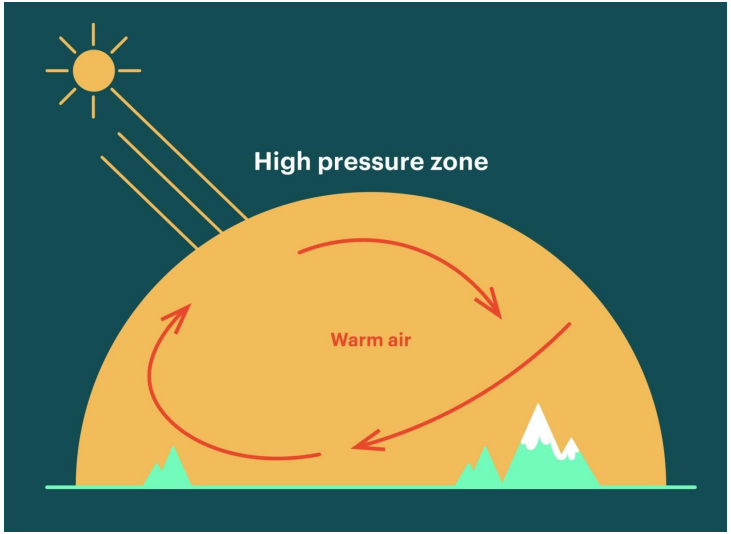


Experimenting

- **Graphing Data**
  - Observing intensity
  - Heat map
- Correlation?
  - Wildfires → smoke
  - High wind speeds



Understanding



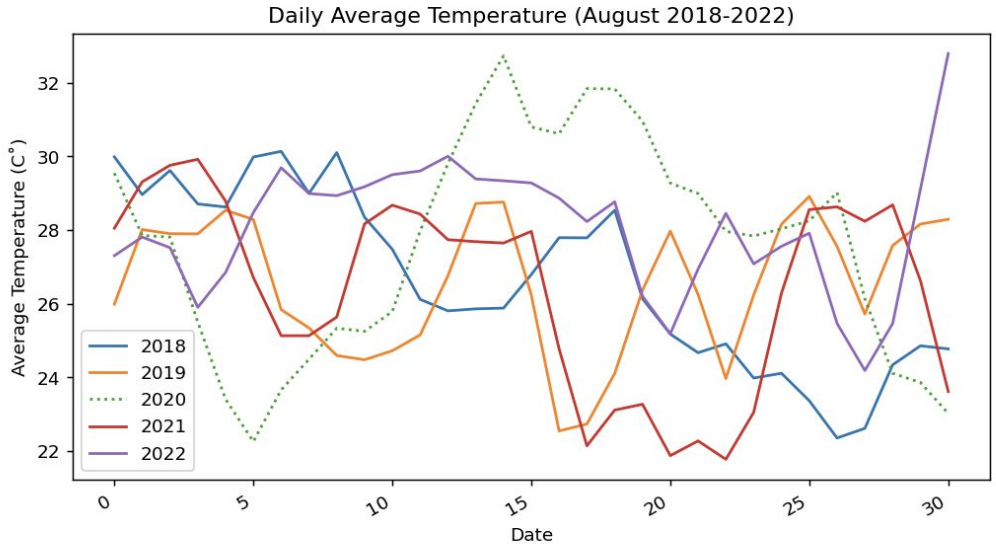
What causes heat waves - Windy.app

- Abnormal heat
  - High air pressure zone “traps” heat close to ground
  - Anticyclone doesn’t allow hot air to rise
  - Lower layer of atmosphere heats up
- Santa Ana dry winds
- Increases chance of wildfires

Experimenting

Comparing Years:

Analyzing the data from the month of August in years 2018-2022, the peaks in the line graph will displays rise of temperatures from the heatwave.



Calculations

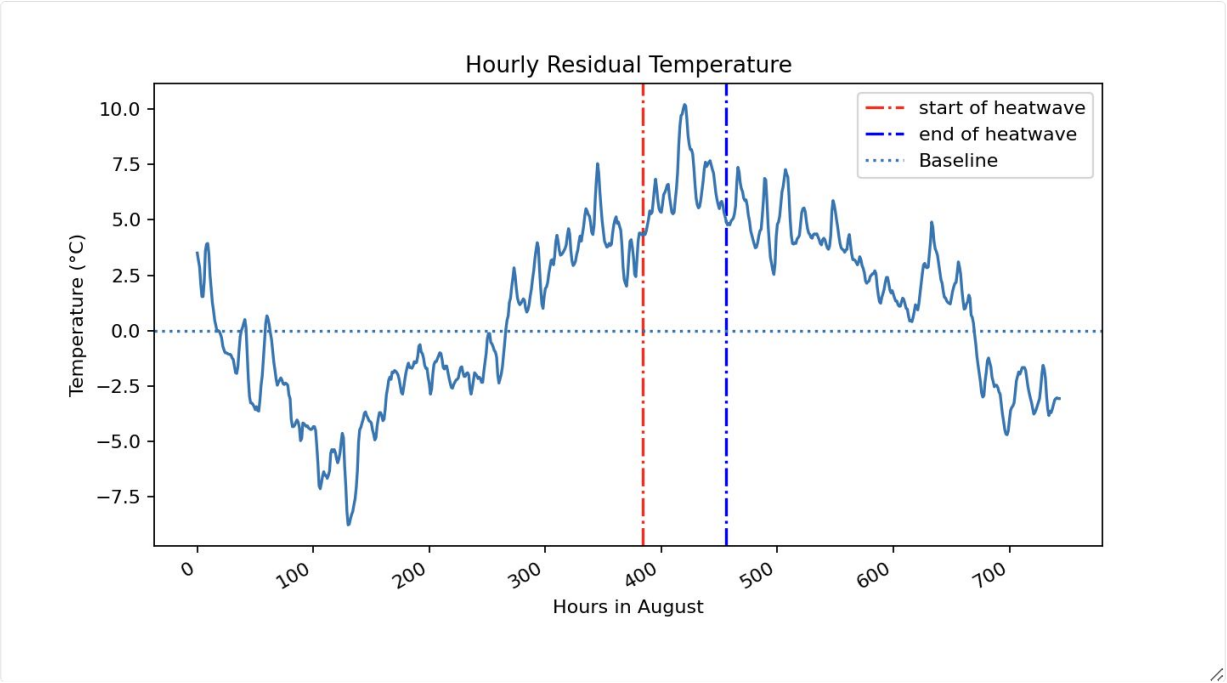
- Measuring magnitude of heatwave
  - Creating a baseline for comparison
  - **Residuals** = actual - median of predicted

(heat wave) (surrounding years)

```python
# Temperature Residual Function
def analyze_baseline(df):
    actual = df.loc['2020-08-01':'2020-08-31'].values
    predicted = np.c_[
        df.loc['2018-08-01':'2018-08-31'].values,
        df.loc['2019-08-01':'2019-08-31'].values,
        df.loc['2021-08-01':'2021-08-31'].values,
        df.loc['2022-08-01':'2022-08-31'].values
    ]
    predicted = np.median(predicted, axis=1)
    return actual - predicted
```

Calculations

- Temperature deviation of August, 2020

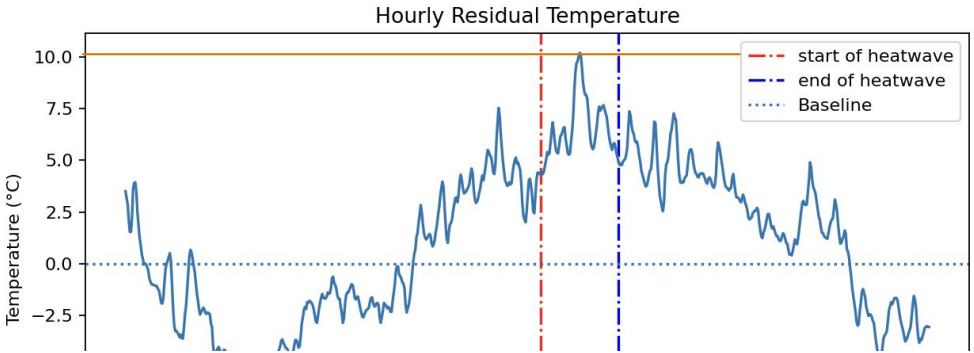


Calculations

- **Max residual** temperature, view peaks

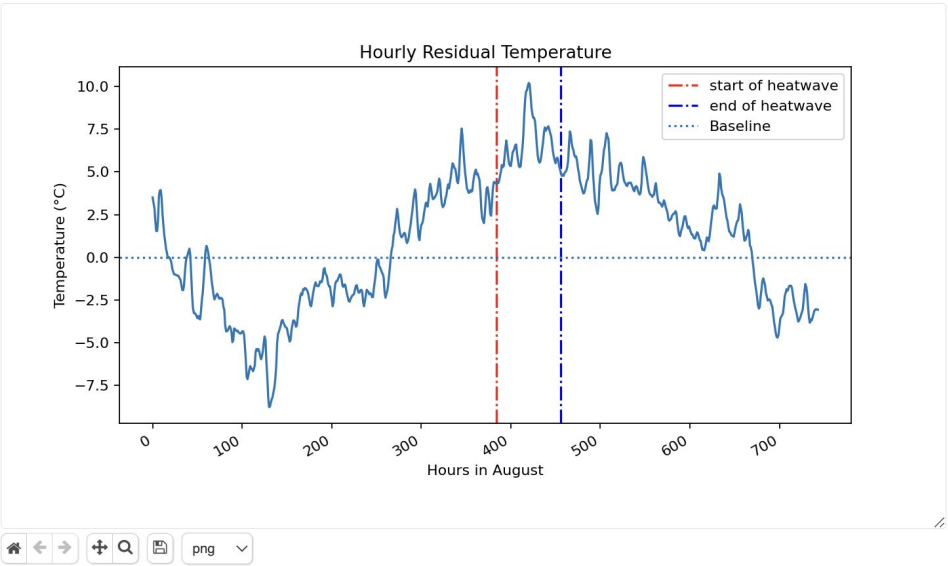
Max residual temperature: 9.80 (C˚)

```
1  max_hourly = hourly_residual.max()
2  mo.md(f"Max residual temperature: {max_hourly:.2f} (C˚)")
```



Calculations

- **Integrals**, area under curve
  - Expecting negative dip 1st half
  - Positive spike 2nd half

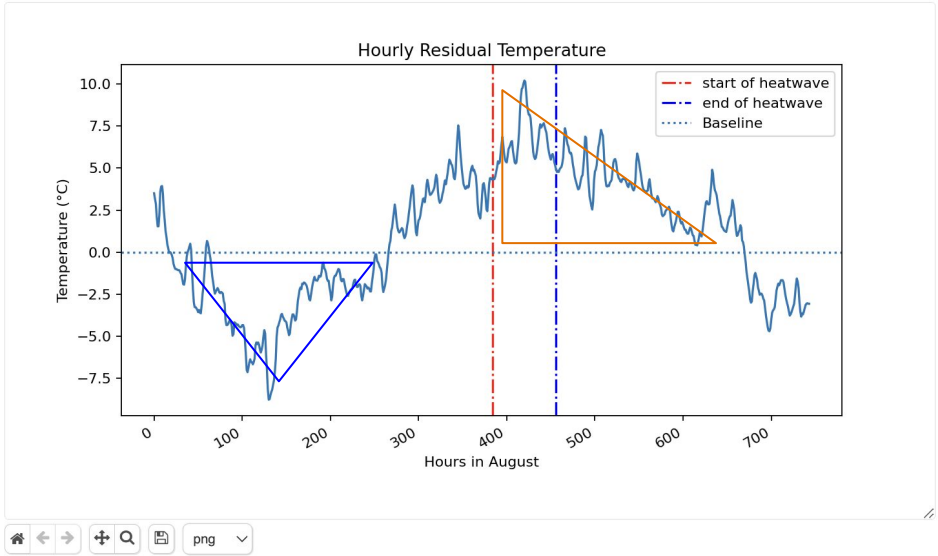


Results

- Max residual temperature: **9.80 (C˚)**
- Overall temperature integral: 25.48 (C˚)
- 1st half: -9.17 (C˚),
- 2nd half: **34.66 (C˚)**

Daily divisions

```
mid_point = len(hourly_residual) // 2
first_integral = np.sum(hourly_residual[:mid_point]) / 24
second_integral = np.sum(hourly_residual[mid_point:]) / 24
hourly_integral = np.sum(hourly_residual) / 24
```



Results

REGROW: Temperature Report

Study of extreme weather across WECC. Report measures the magnitude of the 2020 heatwave through temperature peaks and integrals.

Hourly Statistics

(1) Max residual temperature. (2) August integrals 1st half, 2nd half and overall.

Result clipped. Total rows 126.

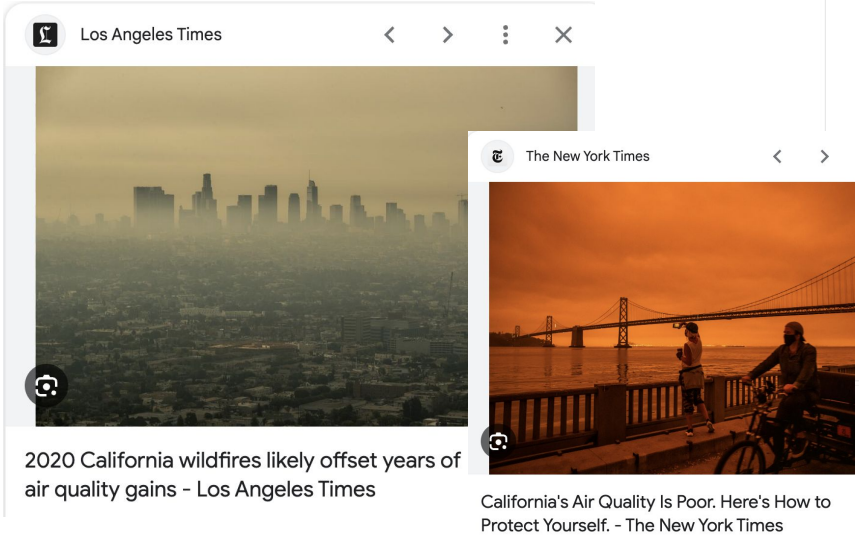
	Max Hourly Residuals	August Hourly Inegrals	First Hourly 1/2 August	Second Hourly 1/2 August
c8h5zm	17.8	82.87	2.69	80.18
c82vu4	14.8	39.42	-22.18	61.6
9rpbwn	8.3	78.98	18.07	60.91
9x0mgn	10.7	85.84	25.33	60.51
9x06gc	9.4	81.59	22.02	59.58

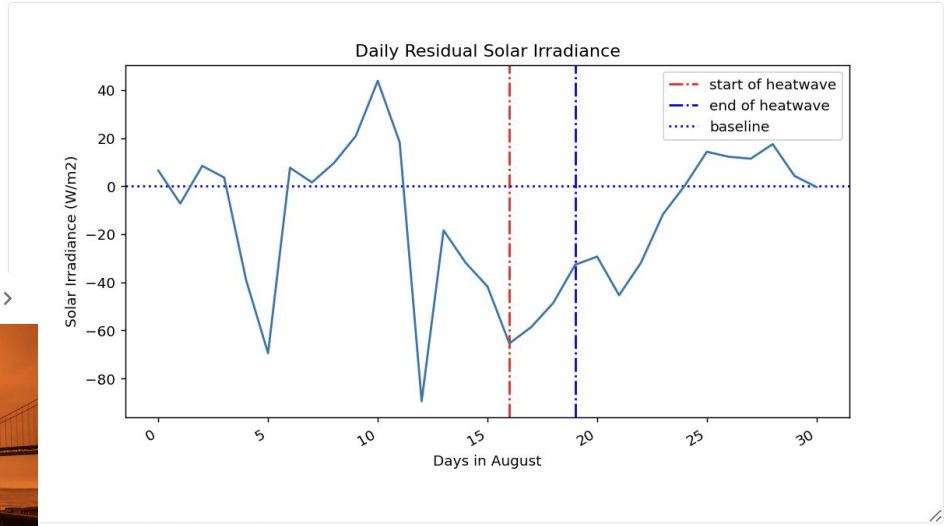
100 rows

Page 1 of 20

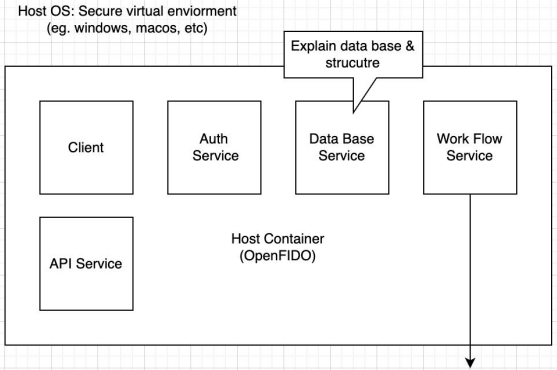
Results

- Loss in solar irradiance





Results - GRIP



OpenFIDO

Augmenting/testing tool used to ingest pole circuit & vegetation data required for simulation

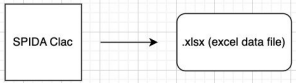
Container - Isolated environment for running an application, faster

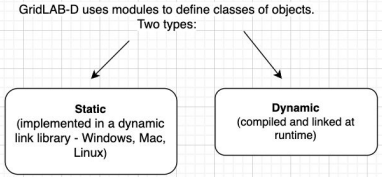
Virtual Environment - Abstraction of a machine, harder to boot up, slower

GridLAB-D

Flexible agent-based simulator that can model the behavior of many objects over time, and keeps track of evolutions.

For GRIP this software tool can be used to support climate adaptation & planning wildfire risk mitigation.

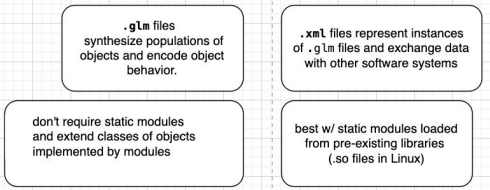




Classes define which properties are allowed in objects, and how behaviors are implemented.

Input Files

GridLAB-D uses two principle kinds of input files.



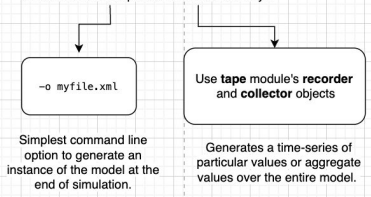
Running GridLAB-D

Simple command line: `gridlabd myfile`

Output is generated to `stdout` and `stderr`

Output Files

GridLAB-D can output result one of two ways:

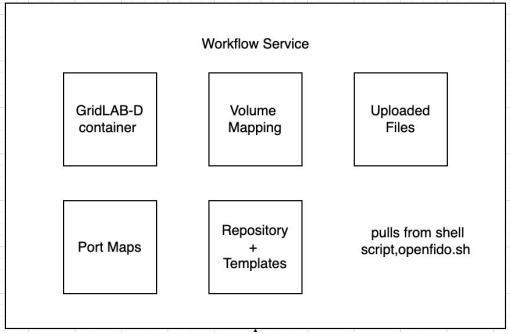


Workflow

1. Circuit model is converted to glm format using Cyme

2. Pole information is extracted from raw SAP Spida prior to processing

Templates - Used for navigating through files & install needed



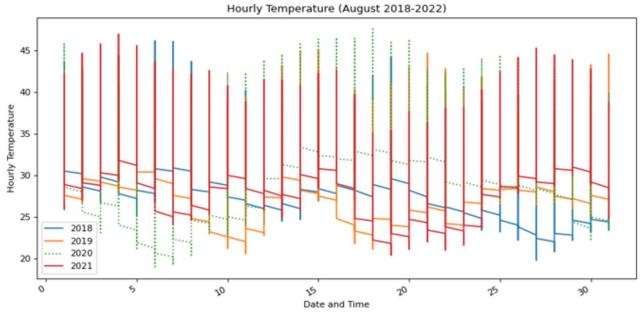
Next Steps

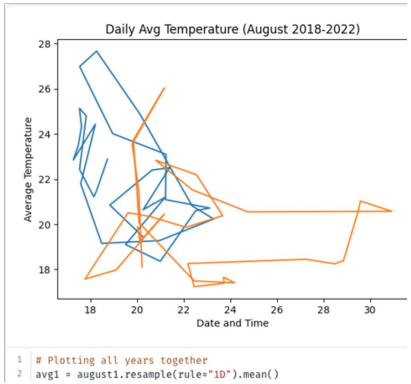


- Locational plotting using geopandas
- Deeper analysis of solar & wind's correlation to extreme events
- Personal Goals
  - Utilize tools like Marimo for visualization and data analysis
  - Application of CS skills
  - Classroom → real world

Conclusion

- Lessons Learned
  - Importance of code review
  - Step by step troubleshooting
  - Version control: Git, terminal
- New skills
  - Python: matplotlib, numpy, pandas
  - *“Documenting observations = science”*
  - Marimo art






```
# Plotting all years together
avg1 = august1.resample(rule="1D").mean()
```

Thank you!
Questions? 😎

Melody Nguyen
Summer 2024

SLAC National Accelerator Laboratory is operated for the US Department of Energy by Stanford University

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